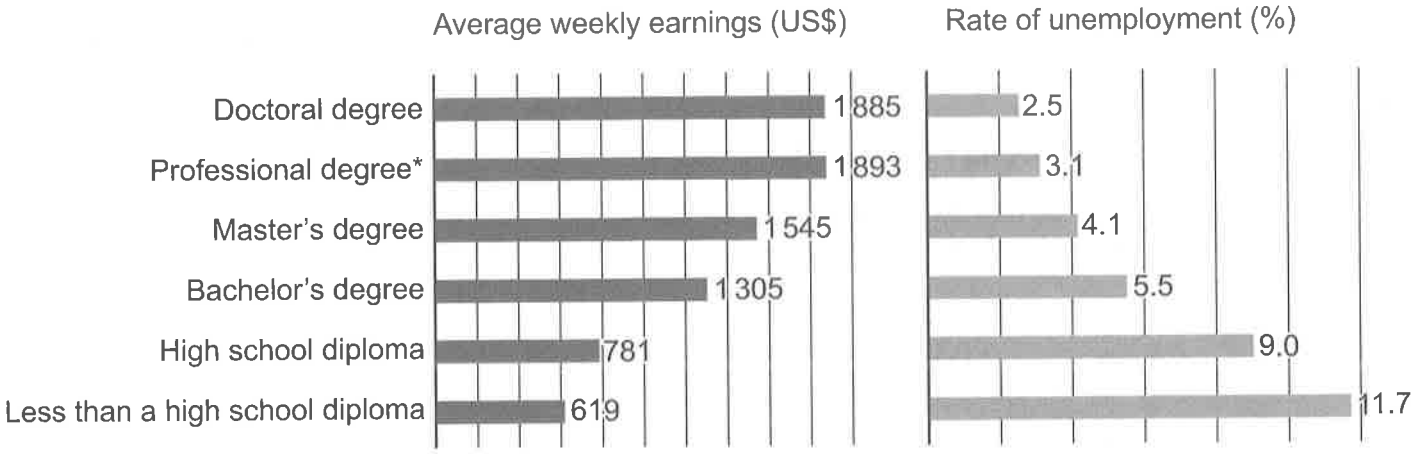
Question 1: Costs and benefits of education

Extract 1: The costs and benefits of Higher Education

Higher Education (HE) refers to the education of students at universities or similar establishments to graduate with a range of different degree level qualifications.

Figure 1: Earnings and unemployment rates by educational attainment, United States (US), 2020



*e.g. pharmacy, medicine, law and teaching

Sources: US Bureau of Labor Statistics and bls.gov, 21 April 2021

Costs and benefits of HE in the United Kingdom (UK)

Between 1970 and 2020, the number of students undertaking HE in the UK increased by over 400%. This increase in HE has to be paid for. Free market economists argue that, as students enjoy additional private benefits from their HE, they should cover the costs. Estimated additional career earnings of graduates compared to non-graduates, after direct taxes and student loan repayments have been paid, was over £252 000 for women and over £168 000 for men. These are the marginal private benefits.

The marginal private cost of HE, including tuition fees and other private costs of education (travel, books etc.), was over £31 000 for a student graduating in 2021. Living costs are not included, as it is argued that young people will incur these whether they go to university or not. Wages forgone must also be included as a private cost. These have been estimated at over £17 000.

It is not only the individual who benefits from HE. Society enjoys an even greater benefit from HE than does the graduate. For example, the extra tax paid by graduates compared with non-graduates is over £300 000 for women and over £260 000 for men. This is a marginal external benefit.

Source: adapted from The Economic Inefficiency of Student fees in England, Intergenerational Foundation, April 2017

Extract 2: The returns to investment made in education

Table 1: Private and social returns to investment in education by per capita income and educational level

	Private return (% per year)			Social return (% per year)		
Countries	Primary education	Secondary education	Higher education	Primary education	Secondary education	Higher education
Low-income	25.4	18.7	26.8	22.1	18.1	13.2
Middle-income	24.5	17.7	20.2	17.1	12.8	11.4
High-income	28.4	13.2	12.8	15.8	10.3	9.7

The private return is the annual rate of return to an individual on the investment made in education. The social return is the annual rate of return in the form of external benefits to society on the investment made in education.

Recent research indicates that the social rates of return in the table are under-estimates of the true returns because of the omission of benefits that cannot be easily expressed in monetary terms, such as reduced mortality rates and lower crime rates. In fact, the social rate of return to investment in education could be 50% higher than the one traditionally estimated.

Source: Returns to Investment in Education: World Bank Group, April 2018

Extract 3: Education and the demand and supply of labour

On average, the difference between the earnings of a university graduate and a high school leaver who does not enter HE is significant in most countries. This difference can, in part, be explained as the increase in productivity that HE gives to individuals. The difference in earnings can also be caused by differences in supply.

A well-trained and educated workforce is likely to be in high demand by employers. Demand will be lower for a workforce that is not well-trained or educated because employers will need to spend a significant amount of time and money training that workforce.

The higher the level of education required for a particular career, the lower the supply of labour. For example, there is a lower supply of medical doctors than nurses because of the additional training required to be a medical doctor.

Source: Principles of Economics, Rice University, September 2021

Extract 4: Educating children

Given the evident benefits of a basic education, why do so many children in so many countries not get one? According to economic theory, the under-provision of a product that is clearly desirable stems from market failure. This can arise from problems on both the demand and supply sides of the market.

The cost to households is one obvious reason why demand for education might be low, given that poor families must first meet their essential needs, such as food and shelter. Education costs could include tuition, books, school supplies and there may be expenses for transport and clothing.

Effective demand for education may be low because of the opportunity costs of educating children. Parents may need their children to work to supplement the household income, or to care for sick or dependent family members. Opportunity costs may make even free schooling unaffordable for some families.

Demand for education is also affected by its perceived value. Parents may not have enough information to assess the private return on an investment in their children's education accurately. They may see the private return as too low to justify the cost, perhaps because of the poor quality of the education available to their children. They may simply be unaware of the opportunities that exist, especially if they are poorly educated themselves.

There may be problems with supply. A government may lack sufficient resources to provide educational services or be administratively incapable of channelling resources to the schools that need them. Government-financed schools may exist in urban areas but not in rural areas or may vary greatly in quality. Such schools may be ranked as a low priority by the country's economic and political leaders. Public resources may be wrongly diverted from education to other projects that serve the interests of economic and political leaders.

Source: Educating Children in Poor Countries, Economic Issues no 33, IMF, 2004

(a) With reference to Figure 1, compare the potential benefits for a graduate in the US with a Bachelor's degree to a school leaver who has a High School Diploma. [2]

(b) With reference to Extract 1 and using a supply and demand diagram, explain **one** possible reason for the higher average earnings of graduates with a Professional degree compared to those with a Bachelor's degree. [3]

(c) 'Opportunity costs may make even free schooling unaffordable for some families.' (Extract 4)

Explain **one** example of an opportunity cost that might make free schooling unaffordable. [3]

(d) Explain how asymmetric information may lead to wrong choices in the market for education. [4]

(e) The government of a low-income country wishes to increase spending on education.

With reference to Table 1, discuss whether the government should concentrate this increase in spending on primary education. [8]

(f) Discuss whether equity issues are more important than market failures as a reason for the government to intervene in the market for education. [10]

[Total: 30]